What Pretraining Doesn't Learn: Current-Value Retrieval Is Latent in Pretrained LLMs

Anonymous ACL submission

Abstract

Language models pretrained on trillions of tokens cannot reliably retrieve the current value of a variable that has been overwritten multiple times in their own input. Across 20 pretrained models (14 open-weight from 270M to 9B parameters, 6 proprietary frontier), this failure manifests as a general loss of position-indexed retrieval: ceiling accuracy on the first value, collapsing accuracy on the current value, and 10–25% accuracy on any intermediate position of the update stream. The failure is not architectural. A 7.4-million-parameter LoRA — 0.24% of one model's weights — trained on 18,000 examples drawn from a small training grid restores accuracy to 93–100% on substantially harder held-out cells, transfers to a different value distribution without modification, and generalises to contexts roughly ten times longer than training; a training-free format change produces similar recovery. Across 82 published checkpoints from two model families, the failure is present at the earliest pretraining checkpoint we can evaluate and survives every documented post-training stage. The capability is present in pretrained models but not surfaced by current pretraining objectives. We argue this is one instance of a broader class of latent capabilities — skills the architecture supports but pretraining does not develop — and present a methodology (characterise, recover, locate mechanistically ⟨**NOTE:** verb-choice pending⟩) for identifying others. ⟨**NOTE:** If §7 lands: add a clause "with recovery localised to specific upper-middle-layer attention heads" before the final sentence.⟩

1 Introduction

Language models are increasingly deployed in settings where the value of a variable changes over time and the model must use the current value, not an earlier one. Multi-turn assistants must respect the most recent user correction; retrieval-augmented systems must use the latest version of a document; tool-augmented agents must read the current contents of a variable after a sequence of writes. Failures of this skill have been documented at the application level — Laban et al. (2025) report substantial degradation in multi-turn dialogues as state updates cascade — but the underlying mechanism, and whether the failure is intrinsic to current models, is not understood. We isolate this skill with a minimal controlled instance of the broader deployment problem.

Stream (K=3, N=3, randomly interleaved):

```
color: red   size: large   shape: round
size: small   color: blue   shape: square
shape: triangle   color: green   size: medium
```

FVQ: What was the *first* value of `color`? → **red**
CVQ: What is the *current* value of `color`? → **green**

Figure 1: Task illustration (K=3, N=3). The stream randomly interleaves updates for all K keys. To answer CVQ correctly, a model must isolate `color` entries from competing keys and identify the most recent one among them.

Our task interleaves K keys each receiving N value updates in a single stream and queries either the first value of a target key (FVQ) or its current value (CVQ); Figure 1 shows a small instance with K=3, N=3. Both queries are trivially solvable in principle: the entire stream is present in context, FVQ asks for the first occurrence of the target key, and CVQ asks for the last. Failure on this task is therefore evidence of a retrieval problem — the model cannot locate the relevant entries in its own input — not a context-length or memory limitation.

We evaluate 20 pretrained models spanning 14 open-weight checkpoints (270M–9B parameters) and 6 proprietary frontier models. Every model exhibits a substantial gap between FVQ and CVQ accuracy in at least one (K, N) cell of our grid (Table 1, Figure 2). At low stream load, models systematically favor the first value (*primacy regime*). In several smaller open-weight models at high load, the gap inverts — current-query accuracy exceeds first-query accuracy — but both remain well below ceiling (*reversal regime*). Neither regime represents correct retrieval; both are deviations from the answer the prompt asks for. When we ask for any intermediate position k of a target key ($1 < k < N$), baseline accuracy collapses to 10–25% (Figure 3; full sweep across six cells in Appendix I), indicating a general loss of position-indexed retrieval rather than a first-vs-current confusion.

The failure is not an architectural limit. A LoRA adapter (Hu et al., 2022) — 0.24% of Qwen2.5-3B-Instruct's parameters, trained on 18,000 examples drawn from a small training grid (against an 18-trillion-token pretraining budget; Qwen Team, 2025) — restores both FVQ and CVQ accuracy to 93–100% on substantially harder held-out cells, well beyond the training grid in both K and N (Table 4). The same LoRA, trained only on one value distribution, transfers to a held-out distribution without modification (Table 5), and recovers the intermediate-position retrieval that the base model lacks (Figure 3). An inference-time format intervention — replacing the plain key:value stream with positionally-anchored alternatives — produces similar recovery without any training (Table 2); this is not the stylistic adjustment small-data alignment achieves (Zhou et al., 2023), but a recovery of a specific mechanistic capability. The recovery is task-specific: a control LoRA trained on arithmetic does not close the gap on update tracking ⟨**NOTE:** pending arithmetic control⟩. Two interventions, one minimal-training and one training-free, converge on the same outcome. The capability is present in the architecture; pretraining does not surface it by default.

We make the following contributions:

- **Behavioral characterization.** We document a robust FVQ–CVQ gap across 20 pretrained models, including a previously-undocumented reversal regime at high load and a general failure of position-indexed retrieval (§4).
- **Two independent recoveries.** We show the failure is recoverable via two independent interventions — a task-specific LoRA and an inference-time format change — demonstrating the capability is latent rather than absent (§5).
- **Training-dynamics analysis.** We trace the failure across 82 pretraining and post-training checkpoints from two model families, showing it is present from the earliest available pretraining step and survives every documented training stage (§6).
- **Mechanism.** We identify the attention-head clusters carrying the bias pre-intervention ⟨**NOTE:** post-LoRA mechanism and pre/post comparison pending⟩ (§7).

Together, these results frame a methodology — **characterise**, **recover**, **locate** — for identifying latent capabilities pretraining does not develop.

2 Related Work

Stateful retrieval failures in LLMs. Laban et al. (2025) report substantial degradation in multi-turn dialogues as state updates cascade and contradict earlier context. Kim and Schuster (2023) and Kim et al. (2024) formalize entity state tracking as updates to a single entity's location or attributes, finding the skill is unevenly present across models. Rezaee et al. (2025) introduce a state-tracking benchmark with three entity-transition tasks, showing that newer-generation models can solve the task with chain-of-thought while older generations fail after a few transitions. The closest direct analogue of our setting is Wang and Sun (2025), who construct a 46-key stream with up to 400 updates and a single current-value query; they find a log-linear accuracy decline and conclude the failure is fundamental. We extend this line in three ways: testing both first-value and current-value queries reveals an asymmetry; dense intermediate-position queries reveal a general failure of position-indexed retrieval; and our interventions demonstrate the failure is recoverable.

Known limits on retrieval in long or complex contexts. Position-bias studies document a U-shaped accuracy curve in static long contexts (Liu et al., 2024) and an "attention sink" on initial tokens regardless of semantic relevance (Xiao et al., 2024); long-context benchmarks such as RULER (Hsieh et al., 2024) and HELMET (Yen et al., 2024) test retrieval of static facts from noise. Theoretical results constrain transformer computation: attention dilutes as sequence length grows (Hahn, 2020), fixed-depth transformers are bounded by first-order logic with majority quantifiers (Merrill and Sabharwal, 2023), and compositional state tracking requires logarithmic depth (Fagnou et al., 2024) — a constraint our destructive-update setting does not invoke. Format sensitivity is also documented: meaning-preserving prompt changes can alter accuracy by up to 76 percentage points (Sclar et al., 2024); our format intervention is a controlled instance. Our task differs from these lines in that updates are *destructive* (earlier values are overwritten, not merely earlier in context) and the bias we observe is regime-dependent, flipping sign with stream load rather than tracing a single U-shape.

Fine-tuning surfaces latent capacity. Recent work has shown that capabilities not visible in pretrained models can be surfaced with minimal supervision. Zhou et al. (2023) demonstrate that 1,000 curated dialogues are sufficient to teach a pretrained model to follow chat-style instructions, attributing this to pretraining having already done the work of knowledge acquisition. Sun and Dredze (2025) make a more general claim across 18 datasets: pretraining improves models in latent ways only visible after fine-tuning. The methods we use — low-rank adaptation (Hu et al., 2022) and the observation that fine-tuning lives in a low intrinsic dimension (Aghajanyan et al., 2021) — are by now standard. Our contribution against this background is the concrete instance with mechanism: a specific capability (in-context update retrieval) that pretrained LLMs systematically fail at, surfaced by an 18,000-example LoRA across held-out grid cells, datasets, and $\sim 10\times$ context lengths, with the recovery traced mechanistically to upper-middle-layer attention heads ⟨**NOTE:** post-LoRA pre/post comparison pending⟩. The distinction from LIMA is style/alignment versus mechanistic capability recovery; the distinction from Amuro & Char is general claim versus concrete instance with mechanism.

Mechanism and pretraining dynamics. Our mechanism and pretraining-dynamics analyses build on two threads. Mechanistic interpretability has identified circuits for specific capabilities — induction heads forming during pretraining as a phase transition (Olsson et al., 2022), indirect-object identification circuits (Wang et al., 2023), automated circuit discovery (Conmy et al., 2023), and interpretable algorithms in successful state trackers (Li et al., 2025). Closest to our setting, Mishra (2025) report that 62% of attention heads shift function during fine-tuning on summarization, with middle layers exhibiting the most dramatic change; we apply the same approach to a mechanistic capability rather than a style task. On the pretraining-capability side, Wei et al. (2022) characterize abilities that appear at scale and Schaeffer et al. (2023) argue these emergences are metric artifacts; our work is a *non-emergence* question — the success of this skill does not emerge at any scale we test. Allen-Zhu and Li (2023) show that knowledge can be memorized during pretraining but not extractable without targeted augmentation, paralleling our latent-capacity claim: the architecture stores the information; pretraining does not develop the retrieval pathway by default.

3 Task and Setup

Task. Let $\mathcal{K} = \{k_1, \ldots, k_K\}$ be a set of keys and let $V_i = (v_{i,1}, \ldots, v_{i,N})$ be an ordered update sequence for key k_i. The input stream σ is a random interleaving of all $K \times N$ pairs $\{(k_i, v_{i,j})\}$, subject to the constraint that no two consecutive entries share the same key. The model receives σ and a target key k_t and is asked one of three queries: **FVQ** (first-value, expected answer $v_{t,1}$); **CVQ** (current-value, expected answer $v_{t,N}$); or **IVQ** (intermediate-value, expected answer $v_{t,k}$ for some $1 < k < N$). Figure 1 shows a small instance. The intermediate-value query is used only in the dense-retrieval analysis (§5; Appendix I).

Datasets. **Arbitrary-Single (ARB)** uses a pool of 2,300 single-token English words across 46 semantic categories; single-token values keep cross-entropy and probing computations clean and make it the dataset for the mechanistic analysis (§7). **Semantic-Multi (SEM)** uses real category members as multi-token values across the same 46 categories (2,403 entries total) and is our primary behavioural dataset — semantic coherence between key and value limits template-matching shortcuts. **Narrative (NAR)** embeds key-value updates in a Dota 2 match commentary; it is used only for the format intervention in §5 ⟨**NOTE:** NAR may be dropped from the final paper if the format intervention story is fully carried by ARB/SEM; decide before submission⟩. Full construction details are in Appendix A.

Grid and trials. We evaluate models on a unified grid $K \in \{2, 3, 5, 7, 10, 15, 20, 25, 30\}$, $N \in \{5, 7, 10, 15, 20, 30, 50, 75, 100\}$, with up to 200 trials per (K, N) cell per query type. Cells where the prompt exceeds a model's context limit or where the dataset pool cannot supply N distinct values per key are excluded (specifics in Appendix A).

Metric. For each (K, N) cell we report FVQ and CVQ accuracy as string-match proportions (matching rule in Appendix B) with Wilson 95% confidence intervals (Wilson, 1927). The primary metric is the **gap** Δ = FVQ accuracy − CVQ accuracy; we call a gap **substantial** when $|\Delta| > 0.05$. All inferences use greedy decoding (temperature 0) to eliminate sampling variance.

Models. We evaluate **11 open-weight models** from three major families — Qwen2.5 (0.5B–3B; instruct and base), Qwen3.5 (0.8B–9B; instruct), Gemma-3 (270M–4B; instruct) — and **6 proprietary frontier models** from OpenAI (GPT-4.1, GPT-4.1-mini), Anthropic (Claude-4.5-Haiku, Claude-Sonnet), and Google (Gemini-2.5-Flash, Gemini-2.5-Pro). Three additional smaller models — TinyLlama-1.1B, StableLM-2-1.6B, Pythia-410M — appear only in the appendix. The full model list is in Appendix C; per-cell behavioural results are in Appendix D.

4 Pretraining Fails at In-Context Update Retrieval

Every model we evaluate exhibits a substantial FVQ–CVQ gap. At the representative cells in Table 1, gaps range from +6 to +53 percentage points, with the majority exceeding +20 pp. Per-cell evidence across the full (K, N) grid for all 20 models is in Appendix D. Figure 2 shows the trajectory in two open-weight models under three axis scans, including one that reaches the reversal regime.

The gap takes two shapes. In most cells of Table 1 — the **primacy regime** — FVQ exceeds CVQ: the model anchors to the first value bound to the key and fails to overwrite. In several smaller open-weight models at higher (K, N), the gap inverts to a **reversal regime**: CVQ exceeds FVQ, but both queries stay well below ceiling, with CVQ accuracy in the 50–65% range. Figure 2's third panel column shows the reversal explicitly for Qwen2.5-3B-Instruct, while Gemma-3-4b-it stays in the primacy regime throughout. Neither regime equals correct retrieval; both are deviations from the answer the prompt asks for. We trace the mechanism underlying the regime split in §7.

Intermediate-position queries $(1 < k < N)$ collapse uniformly across all 5 open-weight models we test under the default Plain format: at $K{=}10, N{=}50$, accuracy drops below 5% within

Model	FVQ	CVQ	Gap
Proprietary models, K=20, N=50			
GPT-4.1	1.00 ± 0.01	0.71 ± 0.07	+0.29
GPT-4.1-mini	1.00 ± 0.01	0.74 ± 0.06	+0.26
Claude-4.5-Haiku	1.00 ± 0.01	0.53 ± 0.07	+0.47
Claude-Sonnet	1.00 ± 0.01	0.74 ± 0.07	+0.26
Gemini-2.5-Flash	1.00 ± 0.01	0.54 ± 0.07	+0.46
Gemini-2.5-Pro	0.90 ± 0.05	0.77 ± 0.06	+0.13
Open-weight models, K=5, N=30			
Qwen2.5-3B-Instruct	0.61 ± 0.09	0.55 ± 0.10	+0.06
Qwen3.5-2B	0.62 ± 0.09	0.48 ± 0.10	+0.14
Qwen3.5-4B	0.86 ± 0.07	0.64 ± 0.09	+0.22
Qwen3.5-9B	0.86 ± 0.07	0.79 ± 0.08	+0.07
Gemma-3-1b-it	0.55 ± 0.10	0.02 ± 0.03	+0.53
Gemma-3-4b-it	0.97 ± 0.04	0.68 ± 0.09	+0.29

Table 1: FVQ and CVQ accuracy on Semantic-Multi at a representative (K, N) cell for each block: $K{=}20, N{=}50$ for proprietary models, $K{=}5, N{=}30$ for open-weight (each is the smallest cell at which every model in the block exhibits gap $> +0.05$). Up to 200 trials per cell; values are accuracy $\pm$ Wilson 95% half-width.

two positions of the stream start and remains there (Appendix F, multi-model IVQ figure, left panel). Three of the six proprietary models we evaluate show the same pattern (GPT-4.1, GPT-4.1-mini, Claude-4.5-Haiku); Claude-Sonnet shows U-shaped recovery and Gemini-2.5-Flash a partial dip; Gemini-2.5-Pro is the single model that maintains near-ceiling accuracy across all positions, indicating the capability is achievable but not developed in the majority of models we test. A deeper per-cell dive for Qwen2.5-3B-Instruct with the LoRA recovery panel is in Appendix I (cited again in §5). The FVQ–CVQ asymmetry measured above is therefore one slice of a broader failure of position-indexed retrieval — present in 8 of 11 models we evaluate.

The rest of the paper asks two questions about this failure: whether it is recoverable from the architecture's existing capacity (§5), and when during pretraining and post-training the failure develops (§6).

5 Two Recoveries: Format and LoRA

5.1 Format intervention

If the failure measured in §4 is about *locating* the relevant entries in the stream, making positions explicit at the surface should make the lookup easier. We test this with four prompt formats applied to the same key-value stream: **Plain** (the default condition used in §4, no positional markers); **Labeled**

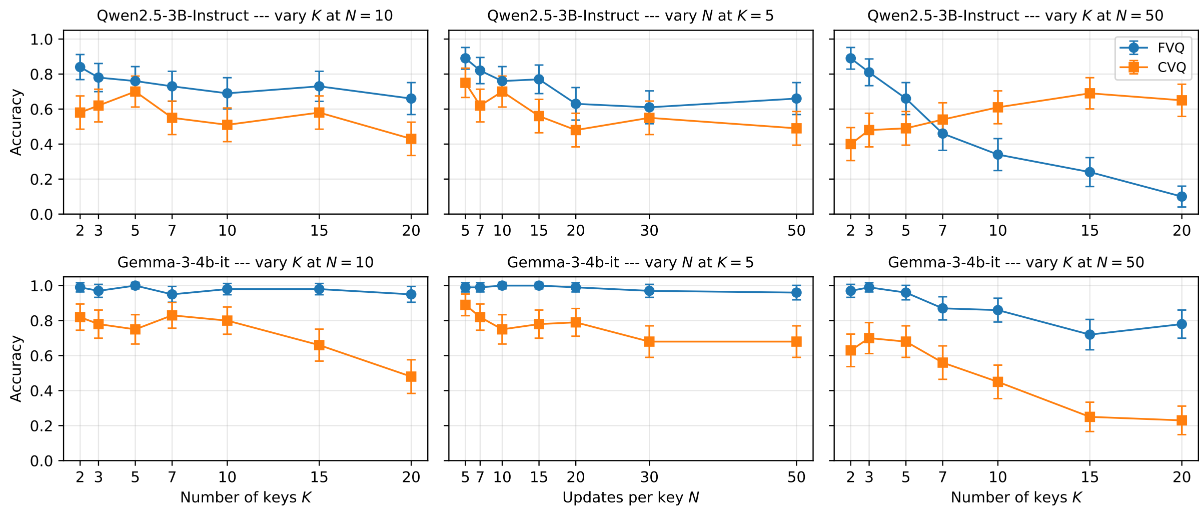

Figure 2: FVQ and CVQ accuracy on Semantic-Multi for two open-weight models. **Top**: Qwen2.5-3B-Instruct. **Bottom**: Gemma-3-4b-it. **Left**: vary K at $N{=}10$ (low-load primacy regime). **Middle**: vary N at $K{=}5$ (low-load primacy regime). **Right**: vary K at $N{=}50$ (high load). The right panel shows a clear crossover into the reversal regime for Qwen2.5-3B-Instruct at $K \approx 7$ (CVQ exceeds FVQ; neither approaches ceiling), while Gemma-3-4b-it stays in the primacy regime throughout. Error bars are Wilson 95% confidence intervals.

Model	Plain	Labeled	Block	Landmark
Open-weight models				
Qwen2.5-3B-Instruct	+0.27	−0.06	−0.07	+0.45
Qwen3.5-2B	+0.94	−0.07	+0.01	+0.73
Qwen3.5-4B	+0.99	−0.04	+0.00	+0.52
Qwen3.5-9B	+0.96	—	—	—
Gemma-3-4b-it	+0.90	−0.09	+0.10	+0.92
Proprietary models				
GPT-4.1	+0.31	+0.04	+0.02	+0.00
GPT-4.1-mini	+0.23	+0.06	+0.00	−0.22
Claude-4.5-Haiku	+0.56	+0.05	+0.01	+0.01
Claude-Sonnet	+0.20	+0.19	+0.00	−0.04
Gemini-2.5-Flash	+0.24	+0.02	+0.00	+0.05
Gemini-2.5-Pro	+0.00	+0.00	+0.00	−0.45

Table 2: FVQ−CVQ gap by format on Semantic-Multi at $K{=}10$, $N{=}50$. Up to 200 trials per (model, format) cell. Plain is the default condition used in §4.

Model	Plain	Labeled	Block	Landmark
Open-weight models				
Qwen2.5-3B-Instruct	0.23	0.39	0.92	0.27
Qwen3.5-2B	0.01	0.93	0.98	0.07
Qwen3.5-4B	0.00	0.97	1.00	0.47
Qwen3.5-9B	0.04	—	—	—
Gemma-3-4b-it	0.01	0.47	0.89	0.00
Proprietary models				
GPT-4.1	0.69	0.96	0.98	0.99
GPT-4.1-mini	0.77	0.94	0.99	1.00
Claude-4.5-Haiku	0.43	0.95	0.99	0.90
Claude-Sonnet	0.80	0.81	1.00	0.99
Gemini-2.5-Flash	0.76	0.98	1.00	0.95
Gemini-2.5-Pro	1.00	1.00	1.00	0.99

Table 3: CVQ accuracy by format on Semantic-Multi at $K{=}10$, $N{=}50$. Under Block, open-weight CVQ rises from ≤ 0.23 (Plain) to ≥ 0.89 for every model; FVQ remains at or near ceiling (Table 2), so the gap closes because both queries are answered correctly.

(each entry tagged with an explicit update index); **Block** (entries grouped under per-round headers); and **Landmark** (block layout with `---` separating rounds, no numbers). Worked examples for each format are in Appendix B. We report results at $K{=}10$, $N{=}50$ — the hardest cell in the format sweep — across 11 models (5 open-weight, 6 proprietary) in Tables 2 (gap) and 3 (CVQ accuracy).

Block format eliminates the FVQ–CVQ gap across every open-weight model where we measure it (Tables 2, 3). The recovery is genuine, not equalisation: CVQ rises from near-zero under Plain to near-ceiling under Block, and FVQ stays at or above its Plain value — both queries are now answered correctly. Proprietary models follow the same pattern at smaller magnitudes; Gemini-2.5-Pro is the only model at ceiling on both queries under Plain, consistent with its non-collapse in §4. ⟨**NOTE:** Qwen3.5-9B has only Plain-format data at $K{=}10$, $N{=}50$; the remaining three formats were not run.⟩

Labeled and **Landmark** close the gap with different consequences. In two open-weight models (Qwen2.5-3B-Instruct, Gemma-3-4b-it) Labeled closes the gap by *degrading FVQ* rather than restoring CVQ — both queries end up around 0.4 in those models (Table 3) and the failure shifts rather

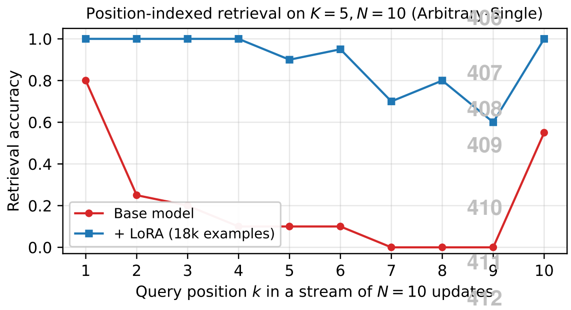

Figure 3: Per-position retrieval accuracy at $K{=}5, N{=}10$ on Arbitrary-Single. The base model recovers position 1 reliably and collapses on intermediate positions; the LoRA-tuned model is at or near ceiling at every queried position.

than disappears. Landmark, which gives round structure without numbers, recovers the proprietary set but leaves the open-weight models near their Plain-format failure (Gemma-3-4b-it gap stays at +0.92). Across the three intervention formats the cleanest reading is that *explicit per-update tokens* — numbers paired with each entry — are what reliably enables retrieval in the open-weight set; visual round structure alone is not enough.

The format intervention demonstrates that the retrieval circuit exists in the pretrained model: given the right surface cue at inference time, the model already has what it needs to retrieve the current value of an in-context variable. Pretraining does not, by default, dispatch this circuit from the Plain key-value format. The same capability is also recoverable from within the model, via targeted fine-tuning, which we show next.

6 When Does the Failure Develop?

⟨**NOTE:** Training dynamics: 82 checkpoints (SmolLM2 + SmolLM3) with stage bands. Two-regime lines per panel. Rules out alignment, SFT, late-pretraining-shift as causes.⟩

7 Mechanism: What LoRA Changes

⟨**NOTE:** Pre-LoRA mechanism (existing data): probing, logit lens, attention routing, per-trial correlation, causal ablation. Post-LoRA mechanism: *pending experiments*. Pre/post comparison: *pending*.⟩

8 Discussion

⟨**NOTE:** Three implications: scaling won't fix this trivially; LM loss has a structural blind spot; characterise–recover–locate as a methodology.⟩

Limitations

⟨**NOTE:** K vs context-length confound; LoRA tested on one or two families; mechanism for one model; no real-world stateful benchmark; reversal-regime predictor absent.⟩

Ethical Considerations

⟨**NOTE:** Standard ACL/EMNLP boilerplate; no human subjects; no sensitive data; synthetic stream construction.⟩

Acknowledgments

⟨**NOTE:** To fill in.⟩

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	Base			**+ LoRA**		
(K, N)	FVQ	CVQ	gap	FVQ	CVQ	gap
$(7, 30)$	50%	38%	+12%	100%	100%	+0%
$(7, 50)$	52%	46%	+6%	100%	96%	+4%
$(10, 30)$	41%	46%	−5%	100%	100%	+0%
$(10, 50)$	42%	50%	−8%	100%	100%	+0%
$(15, 20)$	32%	58%	−26%	100%	98%	+2%
$(15, 30)$	35%	60%	−25%	100%	98%	+2%
$(15, 50)$	18%	59%	−41%	100%	100%	+0%
$(20, 30)$	21%	51%	−30%	100%	98%	+2%
$(20, 50)$	8%	62%	−54%	100%	96%	+4%
$(20, 75)$	6%	56%	−50%	100%	96%	+4%
$(25, 30)$	15%	59%	−44%	100%	98%	+2%
$(25, 50)$	5%	43%	−38%	98%	96%	+2%
$(25, 75)$	6%	60%	−54%	100%	93%	+7%
$(30, 30)$	12%	60%	−48%	100%	98%	+2%
$(30, 50)$	4%	54%	−50%	100%	94%	+6%
$(30, 75)$	1%	56%	−55%	99%	92%	+6%

Table 4: LoRA intervention on Qwen2.5-3B-Instruct evaluated on held-out Arbitrary-Single cells. The adapter is trained on a small grid ($K \leq 10, N \leq 20$) and tested at K up to 30 and N up to 75. Across every held-out cell, both FVQ and CVQ accuracy reach 93–100% post-adapter.

	Base			**+ LoRA (ARB-trained)**		
(K, N)	FVQ	CVQ	gap	FVQ	CVQ	gap
$(7, 20)$	60%	55%	+5%	100%	100%	+0%
$(15, 20)$	25%	55%	−30%	100%	100%	+0%
$(15, 30)$	35%	80%	−45%	100%	100%	+0%
$(20, 30)$	25%	65%	−40%	100%	100%	+0%

Table 5: Cross-distribution transfer: the same LoRA trained only on Arbitrary-Single restores accuracy to ceiling on Semantic-Multi (multi-token semantic values) without modification. Four held-out (K, N) cells; all four go from baseline reversal regime to gap 0%.

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A Dataset Details

We construct three datasets, varying the surface form of the update stream while holding the underlying (K, N) structure fixed. The two key–value variants (ARB and SEM) share schema and grid; the narrative variant (NAR) tests the same task in prose.

Arbitrary-Single (ARB). A pool of 2,300 single-token English words organised into 46 semantic categories (e.g., *color*, *fruit*, *metal*). For each trial, K categories are sampled as keys and N values are drawn without replacement from each category's pool. Keys and values share no inherent update semantics — there is no reason to expect any particular ordering among values of the same category. This isolates tracking from semantic shortcuts and provides the single-token values required for the mechanistic analysis (§7).

Semantic-Multi (SEM). A pool of 2,403 multi-token values across the same 46 categories, where values are real members of each category (e.g., *gemstone* $\rightarrow$ *alexandrite*, *amethyst*, ...). Values are multi-token and drawn from a smaller per-category pool (45–70 values), limiting feasible update counts to $N \leq 50$ for most key counts. SEM is the primary behavioural dataset because the realistic key–value semantics rule out trivial template-matching strategies.

Narrative (NAR). Inspired by Kim and Schuster (2023), who embedded entity state updates in programmatic English sequences, we test whether the same tracking task becomes easier when updates appear in natural prose. We use a Dota 2 match commentary dataset in which K game entities (heroes) each have N attribute updates (e.g., gold, kills) woven into match narration; FVQ and CVQ queries ask for the first and current attribute value of a target hero. ⟨**NOTE:** Re-run GPT-4.1-mini and Claude-Haiku on the matched narrative grid for cross-format comparison; existing runs cover Qwen2.5-3B-Instruct, Qwen3.5-9B, Gemma-3-4b. Decide before submission whether to retain NAR or drop it.⟩

Grid coverage. On the unified grid $K \in \{2, 3, 5, 7, 10, 15, 20, 25, 30\}$, $N \in \{5, 7, 10, 15, 20, 30, 50, 75, 100\}$ (§3), cells where the prompt exceeds a model's context limit are excluded; cells where the dataset pool is insufficient are also excluded (relevant for SEM at high N). This yields up to 79 feasible cells per model on ARB and up to 63 on SEM.

B Prompts, Matching Rule, and Seeds

This appendix documents the exact text used to construct the inference prompts, the four format variants used in §5, the string-matching rule used to score model responses, and the deterministic seeding scheme.

Naming convention. The main text refers to the second query type as *CVQ* (current-value query), reflecting its meaning. The actual prompt text asks for the *last* value of the target key, which is operationally equivalent when the stream is the complete record of updates.

System prompt and queries

System prompt:
You are a precise data extraction tool. Output ONLY a single word — the exact value requested. No other text, no explanation, no punctuation.

Queries (rendered with the target key substituted in):
FVQ: *What was the first value of* [TARGET]*?*
CVQ: *What was the last value of* [TARGET]*?*
IVQ: *What was the* [K]*-th value of* [TARGET]*?*

Figure 4: Inference-time system prompt and per-query phrasing used in all behavioural sweeps reported in the paper. "CVQ" in the main text corresponds to the literal "last value of" query above.

Stream formats. The four formats compared in §5 differ only in how the stream lines are rendered and which user-turn preamble introduces them. Figure 5 shows a small (K=2, N=3) example in each format.

Chat-template rendering. For instruction-tuned models the user/system messages are rendered through the model's tokenizer-default chat template; for base models without a chat template we use a completion-style few-shot wrapper described in the released code.

Matching rule. A model response is counted as correct if, after lowercasing and stripping whitespace, the expected value either equals the response, is contained in the response, or is a prefix of the response. This prefix-tolerant case-insensitive rule accommodates multi-token semantic values where models occasionally emit a trailing qualifier; spot-checks confirm it does not credit lexically unrelated strings.

Seeds. For each $(K, N, \text{query type}, \text{trial index})$ tuple we derive a deterministic seed by hashing the tuple. This ensures the identical stream is presented across base-vs-intervention comparisons at the same cell and trial index, so the two conditions are scored on identical inputs.

C Model Details

Table 6 lists every model evaluated in this paper. The first block is the main open-weight set whose behaviour is analysed in the main text; the second block is three smaller auxiliary models reported in Appendix D for completeness; the third block is the proprietary frontier models. IT = instruction-tuned.

D Full Model Results

⟨**NOTE:** Carry over: per-model SEM grid + proprietary table.⟩

E Arbitrary-Single Replication

⟨**NOTE:** Carry over: ARB grid table.⟩

F Format Intervention Data

This appendix supports two claims made in the main text: the format-intervention results in §5.1 (Tables 2 and 3, both in §5.1) and the multi-model intermediate-position retrieval claim in §4.

Model	**Family**	**Params**	**IT?**
Main open-weight			
Qwen2.5-0.5B-Instruct	Qwen2.5	0.5B	✓
Qwen2.5-1.5B-Instruct	Qwen2.5	1.5B	✓
Qwen2.5-3B-Instruct	Qwen2.5	3B	✓
Qwen2.5-3B	Qwen2.5	3B	×
Qwen3.5-0.8B	Qwen3.5	0.8B	✓
Qwen3.5-2B	Qwen3.5	2B	✓
Qwen3.5-4B	Qwen3.5	4B	✓
Qwen3.5-9B	Qwen3.5	9B	✓
Gemma-3-270m-it	Gemma-3	270M	✓
Gemma-3-1b-it	Gemma-3	1B	✓
Gemma-3-4b-it	Gemma-3	4B	✓
Auxiliary (appendix only)			
TinyLlama-1.1B	LLaMA	1.1B	✓
StableLM-2-1.6B	StableLM	1.6B	✓
Pythia-410M	Pythia	410M	×
Proprietary frontier			
GPT-4.1	OpenAI	—	—
GPT-4.1-mini	OpenAI	—	—
Claude-4.5-Haiku	Anthropic	—	—
Claude-Sonnet	Anthropic	—	—
Gemini-2.5-Flash	Google	—	—
Gemini-2.5-Pro	Google	—	—

Table 6: All 20 models evaluated, grouped by role: 11 main open-weight models that drive the main analyses; 3 smaller auxiliary models reported only in Appendix D; and 6 proprietary frontier models.

Multi-model per-position retrieval under the default Plain format. Figure 6 shows per-position retrieval accuracy at K=10, N=50 on Arbitrary-Single under the default Plain format, separately for the five open-weight models in the ucurve sweep and the six proprietary frontier models. All five open-weight models collapse below 5% accuracy within two positions of the stream start. The proprietary panel reveals three regimes: three models (GPT-4.1, GPT-4.1-mini, Claude-4.5-Haiku) collapse like the open-weight set; Claude-Sonnet and Gemini-2.5-Flash show partial recovery; Gemini-2.5-Pro alone maintains near-ceiling accuracy across all queried positions.

G Training Dynamics Data

⟨**NOTE:** Carry over: SmolLM2 + SmolLM3 trajectory tables.⟩

H LoRA Training Details

⟨**NOTE:** New: LoRA config, training data construction, held-out category and cell split, arithmetic control config, Gemma control config (pending).⟩

I Dense Intermediate-Position Retrieval — Full Sweep

Figure 3 in §5 shows position-by-position retrieval accuracy at a single cell (K=5, N=10). To substantiate the broader claim that the failure is a general loss of position-indexed retrieval, we report the same measurement across all six cells we tested. Figure 7 shows the per-position curves; Table 7 summarises first / intermediate / last position accuracy per cell. Across all six (K, N) cells, baseline intermediate-position accuracy stays at or below 25% and the LoRA-tuned model recovers near-ceiling accuracy at every position queried.

	Base				
(K, N)	Pos. 1	Interm. min–max	Last	Pos. 1	Inter
$(5, 10)$	80%	0%–25%	55%	100%	60
$(10, 20)$	25%	0%–20%	55%	100%	5
$(10, 30)$	65%	0%–10%	5%	100%	0
$(15, 30)$	20%	0%–5%	0%	100%	0
$(20, 30)$	30%	0%–5%	0%	90%	0
$(25, 50)$	5%	0%–5%	0%	90%	0%

Table 7: Per-cell summary of dense intermediate-position retrieval accuracy on Arbitrary-Single. "Interm. min–max" is the range across all intermediate positions ($1 < k < N$) at that cell. The base model loses accuracy at every intermediate position across every cell; the LoRA-tuned model holds near ceiling.

J Pre/Post Mechanism Details

⟨**NOTE:** New: pre/post probing, logit-lens, attention-routing figures and tables (pending experiments).⟩

Plain (`flat_nolabel`; default in §4)

Read the following key-value stream. Each key appears multiple times as it gets updated. Count each occurrence of a key as one update.

```
color: red
size: large
color: blue
size: small
color: green
size: medium
```

What was the 3rd value of color?
Answer with ONLY the exact value. No explanation.

Labeled (`flat_verbose`)

Read the following key-value stream. Each key is updated multiple times.

```
color (update 1): red
size (update 1): large
color (update 2): blue
size (update 2): small
color (update 3): green
size (update 3): medium
```

What was the value of color in Update 3?
Answer with ONLY the exact value. No explanation.

Block (`block`)

Read the following key-value stream. Each key is updated multiple times, grouped by update round.

```
[Update 1]
color: red
size: large
[Update 2]
color: blue
size: small
[Update 3]
color: green
size: medium
```

What was the value of color in Update 3?
Answer with ONLY the exact value. No explanation.

Landmark (`landmark`)

Read the following key-value stream. Each key is updated multiple times, grouped by update round. The '—' marker separates consecutive update rounds.

```
color: red
size: large
—
color: blue
size: small
—
color: green
size: medium
```

What was the value of color in Update 3?
Answer with ONLY the exact value. No explanation.

Figure 5: The four prompt formats compared in §5, illustrated on a K=2, N=3 instance. Same underlying stream content; only the surface structure changes. The "Plain" format is the default for all main-body figures and tables (§4).

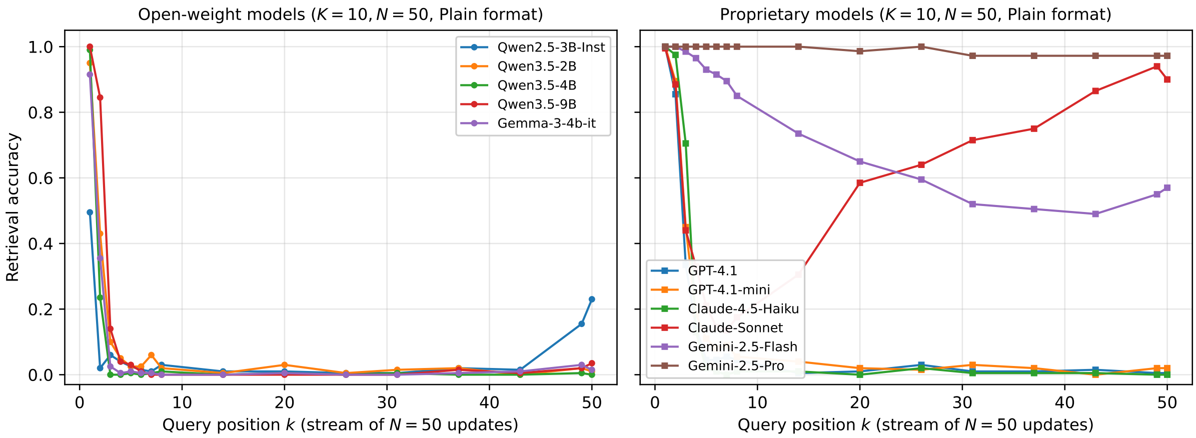

Figure 6: Per-position retrieval accuracy on Arbitrary-Single at $K{=}10, N{=}50$ under the default Plain format (ucurve sweep; `flat_nolabel` condition). Each line is one model. **Left**: five open-weight models, all collapsing uniformly to near-floor by position 2. **Right**: six proprietary models; most collapse but Gemini-2.5-Pro maintains near-ceiling accuracy throughout, and Claude-Sonnet shows U-shaped recovery toward the end of the stream.

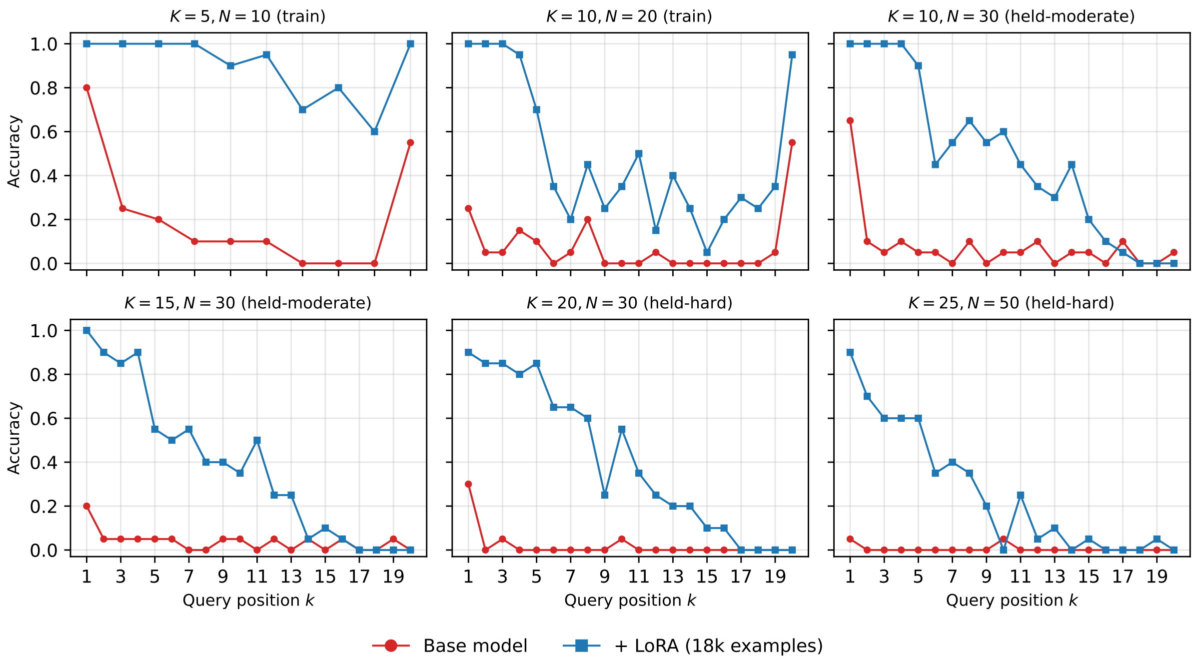

Figure 7: Per-position retrieval accuracy at all six tested (K, N) cells on Arbitrary-Single, base model vs LoRA-tuned model. Group labels (`train`, `held-moderate`, `held-hard`) indicate whether the cell was included in the LoRA training grid.